

L&T EduTech Project Submission

# AI-Based Road Surface Defect Monitoring System

## Day 1: Project Initiation & Data Collection Architecture

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## 1.1 Objective & Problem Definition

The primary objective of Day 1 was to establish the architectural and theoretical foundation for the AI-Based Road Surface Defect Monitoring System. Urban road networks currently rely on manual periodic surveys, which are labor-intensive, prone to human error, and fundamentally lead to incomplete visibility. Defects that manifest between cycles often remain unresolved until they escalate into severe safety hazards. This reactive maintenance paradigm results in traffic disruptions, vehicle damage, and exponentially higher emergency repair costs.

To bridge this gap, our team defined the specifications for an automated, continuous, computer vision-based monitoring system capable of ingesting data from two distinct spatial perspectives: ground-level vehicle dash-cameras and top-down Unmanned Aerial Vehicles (UAVs). This dual-perspective approach guarantees both granular texture analysis and wide-area infrastructural mapping.

## 1.2 Literature Review & Technical Feasibility

Before selecting a dataset, we conducted a brief literature review on current state-of-the-art object detection networks. Traditional two-stage detectors like Faster R-CNN provide high accuracy but fail to meet the real-time inference latency required for moving vehicles processing frames at 30 FPS. Conversely, earlier single-stage models like SSD struggle with small object detection (e.g., minor cracks). We concluded that the YOLOv8 architecture offers the optimal balance of mean Average Precision (mAP) and computational efficiency, making it highly suitable for future edge-deployment on devices like the Raspberry Pi or NVIDIA Jetson.

## 1.3 Dataset Identification & Procurement

Identifying robust datasets capable of generalizing across varied lighting, weather, and geographical conditions was critical to prevent model overfitting.

**Ground-Level Dataset (Vehicle-Mounted):** We procured the Global Road Damage Detection (RDD-2022) dataset, comprising over 47,000 images captured from moving vehicles. This dataset provides diverse geographical representations (India, Japan, USA, Norway, China). This diversity is crucial; asphalt composition, lane markings, and weathering patterns differ drastically globally. The ground-level perspective captures close-up textures critical for identifying mild surface wear and localized depth variations in potholes.

**Aerial Dataset (Drone Survey):** To supplement ground data, we sourced a secondary dataset of high-resolution UAV aerial imagery. The drone perspective offers a wider field-of-view (FOV). While it sacrifices millimeter-level texture detail, it excels at mapping road edges, extensive drainage issues, large-scale cracking grids, and identifying infrastructural

elements like manholes that vehicle cameras might miss due to occlusion by other vehicles.

## 1.4 Defect Categorization Strategy

To ensure the model outputs actionable intelligence rather than just bounding boxes, we established a strict severity-based categorization matrix:

- **D00 (Severe Cracks):** Deep, wide structural failures, often alligator cracking, requiring immediate patching to prevent water ingress and foundation failure.
- **D10 (Mild Cracks):** Superficial surface stress, thermal cracks, or minor longitudinal fissures requiring preventative sealing.
- **D20 (Severe Potholes):** Deep volumetric loss exposing the sub-base, posing immediate hazards to tires, alignment, and suspension.
- **D40 (Mild Potholes):** Shallow surface depressions or minor raveling.
- **D50 (Manholes):** Present specifically in drone imagery. They require explicit classification to prevent the model from registering circular metallic covers as severe potholes.
- **D60 (Smooth Roads):** A critical baseline control class. Supplying the model with negative examples significantly reduces false-positive rates on healthy asphalt.

## 1.5 Anticipated Challenges & Initial Mitigations

During the preliminary data review, we identified several environmental and hardware challenges. Inconsistent illumination (shadows cast by trees or buildings) can mimic the pixel gradients of cracks. Motion blur from vehicle vibration degrades edge sharpness. To mitigate this architecturally, we drafted a preprocessing pipeline involving dynamic speed-adaptive frame dropping and Laplacian variance filters. This ensures blurry or redundant frames are rejected before they consume valuable inference compute, optimizing the pipeline for eventual real-world deployment.